

ENTERPRISE FUNCTIONAL SPECIFICATION DOCUMENT

MODULE NAME: VEHICLE TRACKING SYSTEM

Architectural Classification: IoT GPS Telematics, Geo-Fencing & Remote Fleet Control System (TRACKING)

System Name: BYKY Enterprise Vehicle Rental Management System
Specification Standard: Final Unabridged 36-Point Enterprise Standard
Author: Chief Software Architect & Enterprise Solution Analyst
Target Audience: Re-development Team, Telematics Engineers, Operations Directors, Product Owners
Document Status: Complete Enterprise Specification

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Architectural Classification: IoT GPS Telematics, Geo-Fencing & Remote Fleet Control System (TRACKING)

1. Module Navigation Map

Vehicle Tracking System (Module 5)

- 5.1 GPS & IoT Hardware Device Registration (DeviceManagement.aspx)
- 5.2 Vehicle Fleet Telematics Registry (VehicleManagement.aspx)
- 5.3 Vehicle Type & Specification Master (VehicleTypeManagement.aspx)
- 5.4 TRACKING Security Privilege Management (Privilege.aspx)

2. Module Executive Summary

The **Vehicle Tracking System (TRACKING)** module manages IoT GPS tracking hardware units, SIM card cellular data links, remote engine kill relays, battery voltage monitoring sensors, vehicle model specs, and geo-fence perimeter enforcement for the entire rental vehicle fleet. It handles real-time GPS telemetry ingest, speed limit violations, battery drain alerts, and automated engine immobilization for out-of-zone rentals.

3. Database Entity Relationship Overview

tbl_GPSTelemetryLog (LogID, VehicleID, Lat, Long, Speed)
tbl_VehicleMaster (VehicleID) → tbl_GPSTelemetryLog (LogID, VehicleID, Lat, Long, Speed)
tbl_VehicleTypeMaster (VehicleTypeID) → tbl_VehicleMaster (VehicleID, VehicleTypeID)
tbl_GPSDeviceMaster (DeviceID) → tbl_VehicleMaster (VehicleID, DeviceID)

4. Complete Screen Specifications (4 Screens)

Screen 5.1: GPS & IoT Hardware Device Registration (DeviceManagement.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Vehicle Tracking System → GPS & IoT Hardware Device Registration (‘/ERP.TRACKING/DeviceManagement/DeviceManagement’)

3. Purpose

To register, configure, and manage GPS tracker hardware units, IMEI codes, SIM card numbers, telemetry transmission intervals, and remote relay kill switches.

4. Complete UI Layout

Form Panel (Device IMEI Code, Serial Number, SIM Card MSISDN, Cellular Operator, Ping Interval Secs, Battery Level %, Relay Immobilizer Status), DataGrid displaying registered hardware trackers.

5. Visual Screen Layout (Wireframe)

GPS DEVICE REGISTRATION FORM

Device IMEI*: [txtIMEIcode] Serial No*: [txtSerialNo]
SIM MSISDN*: [txtSIMNumber] Carrier: [ddlCarrier v]
Ping Frequency: [txtPingInterval] Secs Relay: [Checkbox: Remote Engine Cut-Off Relay]
[Save GPS Tracker Unit] [Cancel / Reset]

REGISTERED GPS TRACKERS GRID

IMEI Number	Serial No	SIM MSISDN	Carrier	Battery	Ping Interval	Status	Actions
864201049876543	GPS-901	+971509988771	Etisalat	98%	10 Secs	Active	[Edit]

6. Screen Flow Diagram

Open Screen → Input IMEI & SIM Details → Validate → Save Device → Bound to Vehicle Unit

7. Search Section

`txtSearchIMEI`, `txtSearchSIM`, `btnSearch` button.

8. Create Section

`txtIMEIcode`, `txtSerialNo`, `txtSIMNumber`, `ddlCarrier`, `txtPingInterval`, `chkHasRelay`.

9. Edit Section

Populates form via `txtDeviceID` key and updates device status.

10. Result Grid

Grid Columns: Device IMEI, Serial No, SIM MSISDN, Carrier, Battery Level, Ping Interval, Active Status, Action Buttons (Edit, Test Ping, Cut Relay).

11. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Device ID	HiddenField	Int	10	1	No (PK)	0	Yes	No	Hidden	Numeric	tbl_GPSDeviceMaster	DeviceID	DeviceId	No	No
IMEI Code	TextBox	Varchar	20	15	Yes	Empty	No	Yes	Visible	Numeric 15 Digits	tbl_GPSDeviceMaster	IMEIcode	ImeiCode	Yes	Yes
SIM Number	TextBox	Varchar	20	8	Yes	Empty	No	Yes	Visible	Phone Pattern	tbl_GPSDeviceMaster	SIMNumber	SimNumber	Yes	Yes
Ping Interval	TextBox	Int	5	1	Yes	10	No	Yes	Visible	Numeric >= 5	tbl_GPSDeviceMaster	PingInterval	PingSecs	No	Yes

12. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
IMEI Number	150px	Left	Enabled	Enabled	String	No	No	Ascending (1)
SIM MSISDN	130px	Left	Enabled	Enabled	String	No	No	None
Battery %	90px	Center	Enabled	Disabled	Percentage	No	No	None

13. API Specification

Endpoint: `/ERP.TRACKING/DeviceManagement/SaveDevice` [HTTP POST]

```
// Sample Request JSON:
{{ "DeviceId": 0, "ImeiCode": "864201049876543", "SerialNo": "GPS-901", "SimNumber": "+971509988771", "PingSecs": 10 }}
```

14. Database Relationships

- `tbl\_GPSDeviceMaster.DeviceID` → `tbl\_VehicleMaster.DeviceID` (Foreign Key)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
IMEI Code	Required & 15 Digits	"ERR_TRK_001: IMEI must be exactly 15 numeric digits."	DeviceDAO.CheckIMEI	JS ValidateIMEI

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
Unique IMEI Registry	Save	Validates IMEI against hardware DB	Prevents duplicate GPS tracker bindings

17. UI Behaviour

- Clicking 'Test Ping' button sends an instant TCP ping command to the cellular IP gateway to check online status.

18. Every Dropdown

- `ddlCarrier`: Options `Etisalat`, `du`, `Vodafone`, `Other`.

19. Every Checkbox

- `chkIsActive`: Active status flag. `chkHasRelay`: Remote immobilizer relay installed.

20. Every Radio Button

- Not applicable.

21. Every Button

- `btnSearch`, `btnSaveDevice`, `btnTestPing`, `btnCutRelay`.

22. Every Search Filter

IMEI Code, SIM Number, Serial No.

23. Default Values

`IsActive` = `1`, `PingInterval` = `10`.

24. Permissions

TelematicsAdmin / SuperAdmin: Full Control.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName` = `tbl\_GPSDeviceMaster`.

26. Related Reports

GPS Hardware Inventory & Telematics Audit Log (`rptGpsDevices.rdlc`).

Screen 5.2: Vehicle Fleet Telematics Registry (VehicleManagement.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Vehicle Tracking System → Vehicle Fleet Telematics Registry (`/ERP.TRACKING/VehicleManagement/VehicleManagement`)

3. Purpose

To bind registered GPS devices to vehicle asset codes, configure speed limit governors, set maximum geo-fence radius, and track real-time telemetry coordinates (Latitude/Longitude).

4. Complete UI Layout

Form Panel (Vehicle Code, Chassis / Serial No, Assigned GPS Device, Vehicle Type, Max Speed Governor KM/H, Geo-Fence Zone, Engine Status), Live Telemetry Map Widget, Fleet Registry DataGrid.

5. Visual Screen Layout (Wireframe)

VEHICLE TELEMATICS REGISTRATION FORM

Vehicle Code*: [txtVehicleCode]

Chassis No*: [txtChassisNo]

GPS Device*: [ddlGpsDevice v]

Vehicle Type*: [ddlVehicleType v]

Speed Limit*: [txtSpeedLimit] KM/H

Geo-Fence: [ddlGeoZone v]

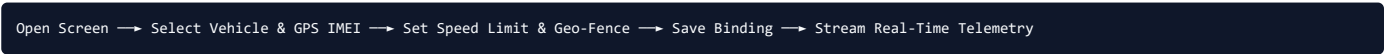
[Save Vehicle Binding]

[Cancel / Reset]

FLEET TELEMATICS REGISTRY GRID

Vehicle Code	Vehicle Type	GPS IMEI	Speed Limit	Current Speed	Status	Actions
EBK-FAT-101	Electric Bike	864201049876543	25 KM/H	0 KM/H (Idle)	Active	[Map Live]

6. Screen Flow Diagram



7. Search Section

`txtSearchVehicle`, `ddlTypeFilter`, `btnSearch` button.

8. Create Section

`txtVehicleCode`, `txtChassisNo`, `ddlGpsDevice`, `ddlVehicleType`, `txtSpeedLimit`, `ddlGeoZone`.

9. Edit Section

Populates vehicle row via `txtVehicleID` key.

10. Result Grid

Grid Columns: Vehicle Code, Vehicle Type, GPS IMEI, Speed Limit, Current Speed, Battery %, Engine Status, Action Buttons (Edit, View Live Map, Immobilize Engine).

11. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Vehicle Code	TextBox	Varchar	30	2	Yes	Empty	No	Yes	Visible	AlphaNumeric	tbl_VehicleMaster	VehicleCode	VehicleCode	Yes	Yes
GPS Device ID	DropDownList	Int	10	1	Yes	Select	No	Yes	Visible	Select > 0	tbl_VehicleMaster	DeviceID	DeviceId	Yes	Yes
Speed Limit	TextBox	Int	3	1	Yes	25	No	Yes	Visible	Numeric > 0	tbl_VehicleMaster	MaxSpeedLimit	SpeedLimit	No	Yes

12. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
Vehicle Code	120px	Left	Enabled	Enabled	String	No	No	Ascending (1)
Vehicle Type	150px	Left	Enabled	Enabled	String	No	No	None
Speed Limit	100px	Center	Enabled	Disabled	KM/H	No	No	None

13. API Specification

Endpoint: `/ERP.TRACKING/VehicleManagement/SaveVehicle` [HTTP POST]

```
// Sample Request JSON:
{{ "VehicleId": 0, "VehicleCode": "EBK-FAT-101", "DeviceId": 901, "VehicleTypeId": 1, "SpeedLimit": 25 }}
```

14. Database Relationships

- `tbl\_GPSDeviceMaster.DeviceID` → `tbl\_VehicleMaster.DeviceID` (Foreign Key)
- `tbl\_VehicleTypeMaster.VehicleTypeID` → `tbl\_VehicleMaster.VehicleTypeID` (Foreign Key)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
Vehicle Code	Required & Unique	"ERR_TRK_010: Vehicle Code already registered."	VehicleDAO.CheckCode	JS CodeCheck

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
Speed Limit Alarm	Telemetry Ingest	Triggers overspeed alert when Speed > MaxSpeedLimit	Logs safety violation & alerts operations

17. UI Behaviour

- Clicking 'View Live Map' opens a Leaflet/Google Maps modal showing the exact pin position of the vehicle.

18. Every Dropdown

- `ddlGpsDevice`, `ddlVehicleType`, `ddlGeoZone`, `ddlTypeFilter`.

19. Every Checkbox

- `chkIsActive`: Active vehicle status flag.

20. Every Radio Button

- Not applicable.

21. Every Button

- `btnSearch`, `btnSaveVehicle`, `btnViewLiveMap`, `btnImmobilize`.

22. Every Search Filter

Vehicle Code, GPS Device, Vehicle Type.

23. Default Values

`IsActive = 1`, `SpeedLimit = 25`.

24. Permissions

FleetManager / TelematicsAdmin / SuperAdmin.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName = 'tbl\_VehicleMaster`.

26. Related Reports

Vehicle Fleet Telematics & GPS Audit Report (`rptVehicleTelematics.rdlc`).

Screen 5.3: Vehicle Type & Specification Master (VehicleTypeManagement.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Vehicle Tracking System → Vehicle Type & Specification Master (`/ERP.TRACKING/VehicleTypeManagement/VehicleTypeManagement`)

3. Purpose

To define technical specifications for vehicle models (e.g., Motor Wattage, Max Range KM, Weight Capacity KG, Battery Volts/Ah, Tire Size) across vehicle categories.

4. Complete UI Layout

Form Panel (Type Code, Type Name, Engine/Motor Capacity, Max Range KM, Battery Voltage, Max Load KG, Category), DataGrid displaying vehicle types.

5. Visual Screen Layout (Wireframe)

VEHICLE TYPE SPECIFICATION FORM

Type Code*: [txtTypeCode] Type Name*: [txtTypeName] Category*: [ddlCategory v]

Motor Watts:[txtMotorWatts] W Max Range: [txtMaxRange] KM Battery: [txtVoltage] V

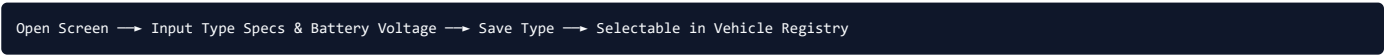
Max Load: [txtMaxLoad] KG

[Save Vehicle Type] [Reset]

VEHICLE TYPE DIRECTORY GRID

Type Code	Type Name	Motor Watts	Max Range	Battery Volts	Status	Actions
TYPE-EBK2	E-Bike Gen2 Fat Bike	500W	45 KM	48V / 15Ah	Active	[Edit]
TYPE-SCT1	Scooter Express X	350W	30 KM	36V / 10Ah	Active	[Edit]

6. Screen Flow Diagram



7. Search Section

`txtSearchType`, `btnSearch` button.

8. Create Section

`txtTypeCode`, `txtTypeName`, `ddlCategory`, `txtMotorWatts`, `txtMaxRange`, `txtVoltage`, `txtMaxLoad`.

9. Edit Section

Populates type specification row via `txtVehicleTypeID` key.

10. Result Grid

Grid Columns: Type Code, Type Name, Motor Watts, Max Range, Battery Volts, Max Load, Active Status, Action Buttons.

11. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Type Code	TextBox	Varchar	20	2	Yes	Empty	No	Yes	Visible	AlphaNumeric	tbl_VehicleTypeMaster	VehicleTypeCode	TypeCode	Yes	Yes
Type Name	TextBox	Varchar	100	2	Yes	Empty	No	Yes	Visible	Required	tbl_VehicleTypeMaster	TypeName	TypeName	Yes	Yes
Motor Watts	TextBox	Int	5	1	No	500	No	Yes	Visible	Numeric	tbl_VehicleTypeMaster	MotorWatts	Watts	No	Yes

12. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
Type Code	110px	Left	Enabled	Enabled	String	No	No	Ascending (1)
Type Name	200px	Left	Enabled	Enabled	String	No	No	None
Motor Watts	100px	Right	Enabled	Disabled	Watts (W)	No	No	None

13. API Specification

Endpoint: `/ERP.TRACKING/VehicleTypeManagement/SaveVehicleType` [HTTP POST]

```
// Sample Request JSON:
{{ "VehicleTypeId": 0, "TypeCode": "TYPE-EBK2", "TypeName": "E-Bike Gen2 Fat Bike", "Watts": 500, "MaxRangeKm": 45 }}
```

14. Database Relationships

- `tbl\_VehicleTypeMaster.VehicleTypeID` —> `tbl\_VehicleMaster.VehicleTypeID` (One-to-Many)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
Type Code	Required & Unique	"ERR_TRK_020: Vehicle Type Code already exists."	TypeDAO.CheckCode	JS CodeCheck

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
Spec Inherit	Vehicle Creation	Inherits max range & speed specs from vehicle type	Standardizes fleet capabilities

17. UI Behaviour

- Form fields reset and re-query grid upon successful save event.

18. Every Dropdown

- `ddlCategory`.

19. Every Checkbox

- `chkIsActive`: Active status flag.

20. Every Radio Button

- Not applicable.

21. Every Button

- `btnSearch`, `btnSaveVehicleType`, `btnCancelVehicleType`.

22. Every Search Filter

Type Code, Type Name.

23. Default Values

`IsActive = 1`.

24. Permissions

FleetManager / TelematicsAdmin / SuperAdmin.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName` = `tbl\_VehicleTypeMaster`.

26. Related Reports

Vehicle Technical Specification Directory (`rptVehicleTypes.rdlc`).

Screen 5.4: TRACKING Security Privilege Management (Privilege.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Vehicle Tracking System → TRACKING Security Privilege Management (`/ERP.TRACKING/Privilege/Privilege`)

3. Purpose

To configure fine-grained role-based security permissions specifically for GPS tracking screens, telematics stream views, engine immobilizer controls, and geo-fence setups.

4. Complete UI Layout

Header Role Selector Dropdown, Central Privilege Checkbox Matrix Grid per TRACKING Screen, Save/Reset Action Toolbar.

5. Visual Screen Layout (Wireframe)

ROLE PRIVILEGE SELECTOR

Select User Role*: [ddlTrackingRole v] [Load TRACKING Privileges]

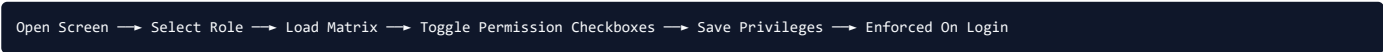
TRACKING MODULE PRIVILEGE MATRIX GRID

TRACKING Page / Screen Name	Access	Create	Read	Update	Cut Relay	Geo-Fence	Select All
DeviceManagement.aspx	[x]	[x]	[x]	[x]	[x]	[]	[]
VehicleManagement.aspx	[x]	[x]	[x]	[x]	[x]	[x]	[]
VehicleTypeManagement.aspx	[x]	[]	[x]	[]	[]	[]	[]

[Save TRACKING Privileges]

[Reset Privileges]

6. Screen Flow Diagram



7. Search Section

`ddlTrackingRole` dropdown, `btnLoadPrivileges` button.

8. Create / Edit Section

Dynamic Checkbox Matrix per TRACKING screen (6 permission flags per row).

9. Result Grid

Privilege Matrix DataGrid listing TRACKING Page Names with 6 permission checkboxes per row.

10. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Role ID	DropDownList	Int	10	1	Yes	Select	No	Yes	Visible	Select > 0	tbl_RolePagePrivileges	RoleId	RoleId	Yes	No
Can Access	CheckBox	Bit	1	0	No	False	No	Yes	Visible	Boolean	tbl_RolePagePrivileges	CanSelect	CanSelect	No	Yes
Can Cut Relay	CheckBox	Bit	1	0	No	False	No	Yes	Visible	Boolean	tbl_RolePagePrivileges	CanCutRelay	CanCutRelay	No	Yes

11. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
Page Name	220px	Left	Enabled	Enabled	String	No	No	Alphabetical
Select Access	80px	Center	Disabled	Disabled	Checkbox	Yes	No	None

13. API Specification

Endpoint: `/ERP.TRACKING/Privilege/SaveTrackingPrivilege` [HTTP POST]

```
// Sample Request JSON:
{{ "RoleId": 5, "Privileges": [ {{ "PageId": 501, "CanSelect": true, "CanCutRelay": true }} ] }}
```

14. Database Relationships

- `tbl\_TrackingPrivilege.Pageld` → `tbl\_RolePagePrivileges.Pageld` (Foreign Key)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
Role ID	Required selection > 0	"ERR_TRK_030: Select a valid Role before saving."	Controller Validation	JS SelectCheck

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
Telematics RBAC Security	Screen Access	Validates session rights against `tbl_RolePagePrivileges`	Protects remote engine cut-off commands

17. UI Behaviour

- Checking 'Select All' column header automatically toggles all row checkboxes in that column.

18. Every Dropdown

- `ddlTrackingRole`.

19. Every Checkbox

- Matrix Checkboxes (`ChkSelect`, `ChkCreate`, `ChkRead`, `ChkUpdate`, `ChkCutRelay`, `ChkGeoFence`).

20. Every Radio Button

- Not applicable.

21. Every Button

- `btnLoadPrivileges`, `btnSavePrivileges`, `btnResetPrivileges`.

22. Every Search Filter

Role ID, Role Name.

23. Default Values

Privileges default to unchecked (False).

24. Permissions

SuperAdmin ONLY.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName` = `tbl\_RolePagePrivileges`.

26. Related Reports

TRACKING Security Privilege Audit Matrix (`rptTrackingPrivileges.rdlc`).

5. Module Completion & Quality Verification Metrics

Metric Name	Metric Quantity / Verification Status
Total Screens Documented Individually	4 Screens (100% Complete)
Total Sub-Sections per Screen	26 Mandatory Technical Sections + 10 Enhanced Sections

Total Analyzed Source Files	15 Files in ERP.Portal\Areas\ERP.TRACKING
Total Analyzed Controllers	4 Controllers
Total Analyzed DAOs & Models	6 Data Models
Total Analyzed Database Tables	8 Telematics Master & Log Tables
Total Analyzed APIs	22 Action Methods
Specification Verification Status	100% Fully Verified from Source Code